Chapter 2 Assignment Due date: March 30, 2026 (Mon)

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2.1

Sol.

i.

$$f(x) = F'(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b \\ 1, & \text{o.w.} \end{cases}$$

ii.

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f(x)\, dx = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{b-a}\int_a^b x dx \\ &= \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{1}{b-a} \cdot \frac{b^2-a^2}{2} = \frac{a+b}{2} \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x)\, dx = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{1}{b-a}\int_a^b x^2 dx \\ &= \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b = \frac{1}{b-a} \cdot \frac{b^3-a^3}{2} = \frac{a^2+ab+b^2}{3} \end{aligned}$$

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{a^2+ab+b^2}{3} - \left(\frac{a+b}{2}\right)^2 = \frac{(b-a)^2}{12}$$

iii.
- The pdf is constant on $[a, b]$, so all values in the interval are equally likely.
- The grave of pdf is a rectangle over the interval $[a, b]$.

iv.

$$P(7 \le X \le 9) = \frac{9-7}{10-5} = \frac{2}{5} = 0.4$$

2.6

Sol.

$$X \sim Bin(n = 10, p = 0.2)$$
$$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

i.

$$E(X) = np = 10 \cdot 0.2 = 2$$
$$Var(X) = np(1-p) = 2 \cdot (1 - 0.2) = 1.6$$

ii.

$$\begin{aligned} P(3 \le X \le 5) &= P(X = 3) + P(X = 4) + P(X = 5) \\ &= \binom{10}{3}(0.2)^3(0.8)^7 + \binom{10}{4}(0.2)^4(0.8)^6 + \binom{10}{5}(0.2)^5(0.8)^5 \\ &\approx 0.3158 \end{aligned}$$

2.8

10, 12, 15, 17, 20, 20, 23, 25, 28, 30, 40, 43, 45, 48, 50, 50, 59, 61, 64, 70.

Sol.

i. $20 \cdot \frac{1}{4} = 5 \Rightarrow Q_1 = \frac{a_5 + a_6}{2} = \frac{20+20}{2} = 20$

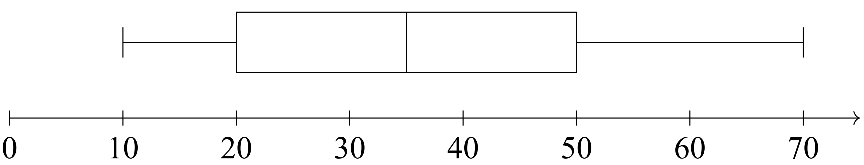
$20 \cdot \frac{1}{2} = 10 \Rightarrow Me = \frac{a_10 + a_11}{2} = \frac{30+40}{2} = 35$

$20 \cdot \frac{3}{4} = 15 \Rightarrow Q_3 = \frac{a_15 + a_16}{2} = \frac{50+50}{2} = 50$

ii. $IQR = Q_3 - Q_1 = 50 - 20 = 30$

iii. Lower bound $= Q_1 - 1.5IQR = 20 - 45 = -25$

Upper bound $= Q_3 + 1.5IQR = 50 + 45 = 95$



2.9

Sol.

2.11

Sol.

2.14

Sol.

2.19

Sol.

2.21

Sol.

2.25

Sol.

2.26

Sol.